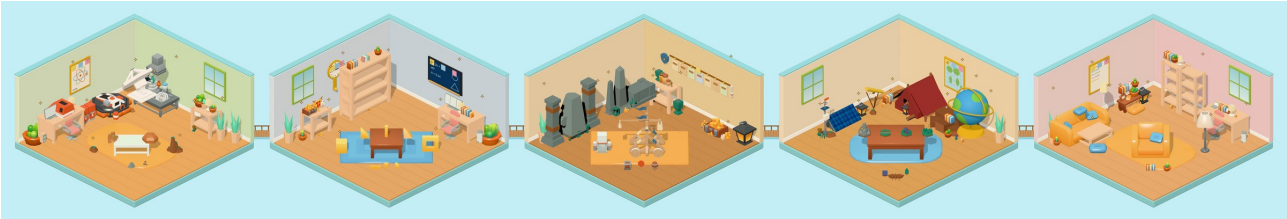
Design

The house, the rooms, every upgradable item and every level, with the art for each. Written so a developer or an AI coding assistant can build the house part of the web app from this document alone.



The Cottage (tier 1), every item at level 10.

Contents

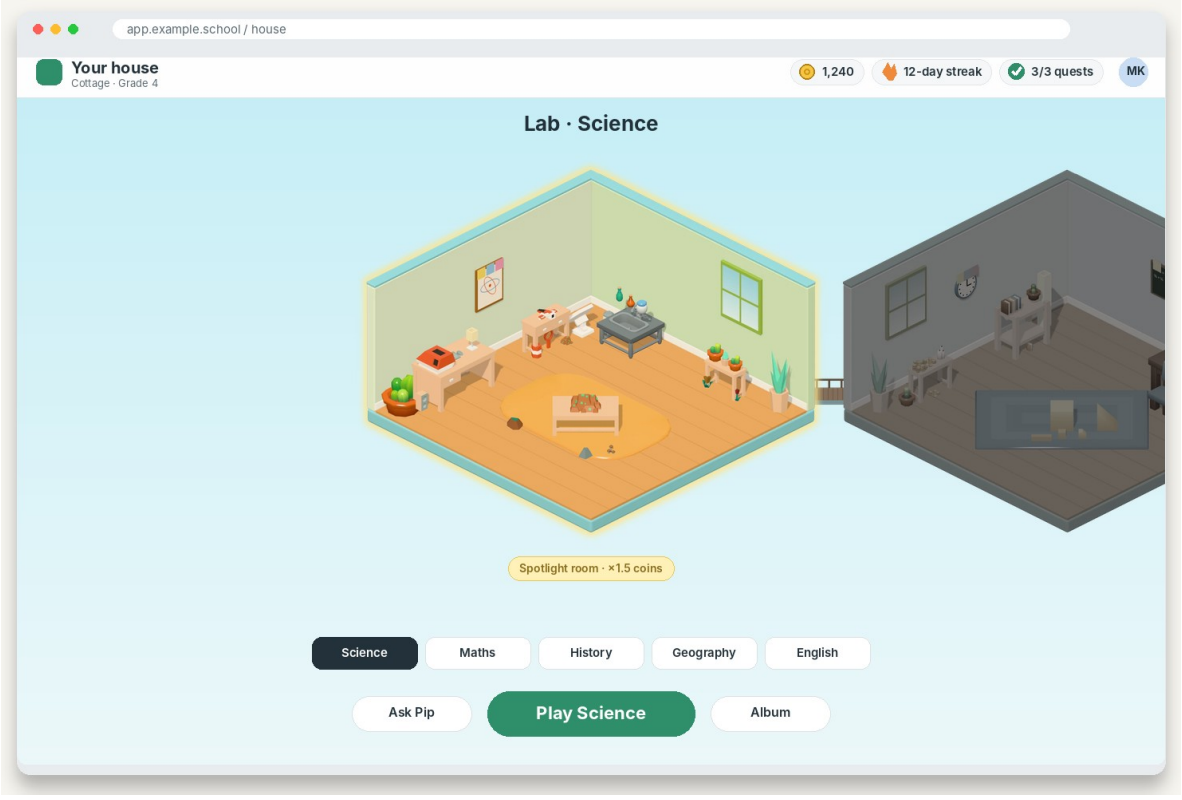
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1. App and room overview

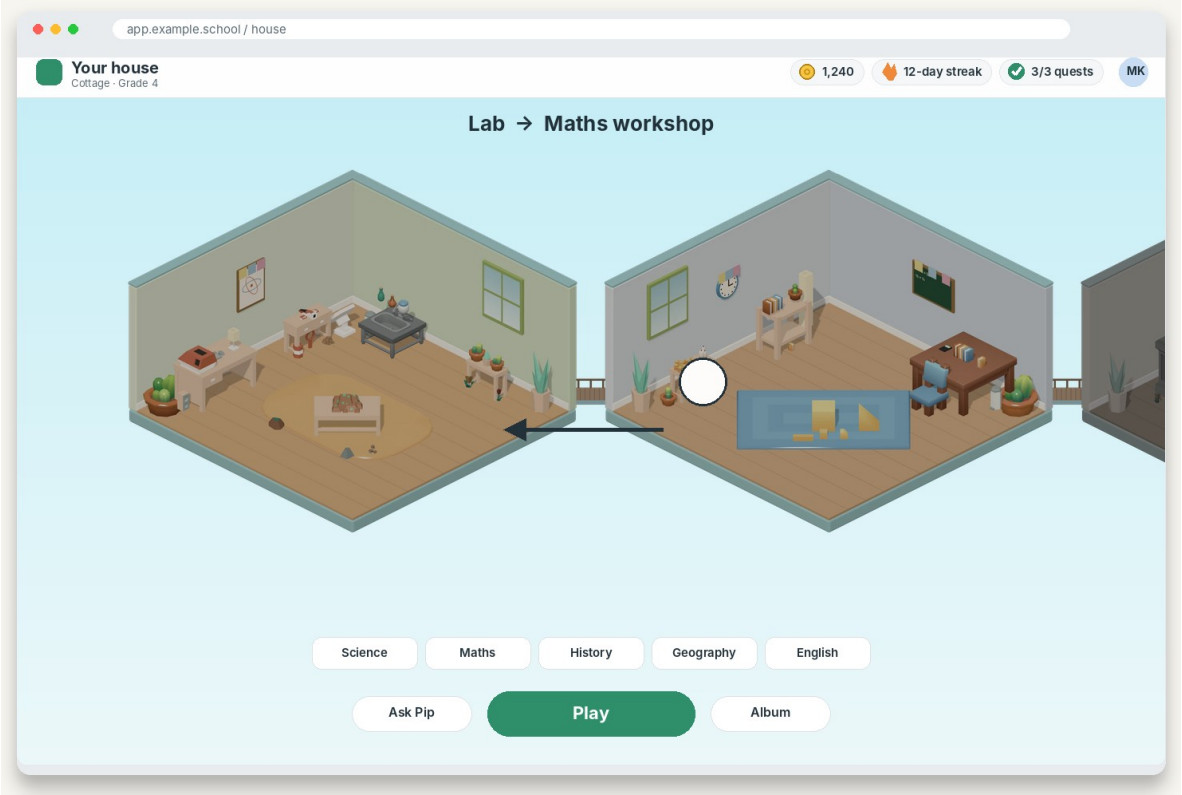
Students learn by playing short games. Every game pays coins. Coins upgrade objects in their own house. The house has five subject rooms joined in a row. A room's objects get visibly better as the student plays, and when every object reaches level 10 the student moves to a bigger house (the next tier). This document covers the house side of the web app: pages, layout, items, levels, art and how to build it. Games, AI and the backend are in the separate build plan.

Thing	Value
Platform	Web app: React + TypeScript + Vite, installable (PWA), works with mouse and touch. FastAPI backend.
Art	Free CC0 3D models (KayKit, Kenney) rendered to images ahead of time. The browser only places images; no 3D code at runtime.
Tiers in this build	Tier 1 Cottage (Grade 4, 6 × 6 m rooms) and Tier 2 Big House (Grade 5, 7 × 7 m rooms). Apartment and Mansion are shown locked.
Rooms per house	5, always in this order: Lab (Science) · Maths workshop · Museum (History) · Explorer's den (Geography) · Reading nook (English).
Upgradable items	6 per room (5 on the floor, 1 on a wall) = 30 per tier, 60 in total.
Levels per item	10. Stages at levels 1, 4, 7, 10; add-ons at 2, 3, 5; glow at 6, gold rim at 8, sparkles at 9.
Cost	Level n costs 20 × n coins (level 1 free). One item 1 → 10 = 1,080 coins. One house = 32,400 coins.
Mastery gate	Levels 8, 9 and 10 also need 70% average mastery in that room's subject.
Tier-up	All 30 items at level 10 → next tier. A teacher can also promote or demote a student (logged with a reason).

1.1 Home page: the connected house



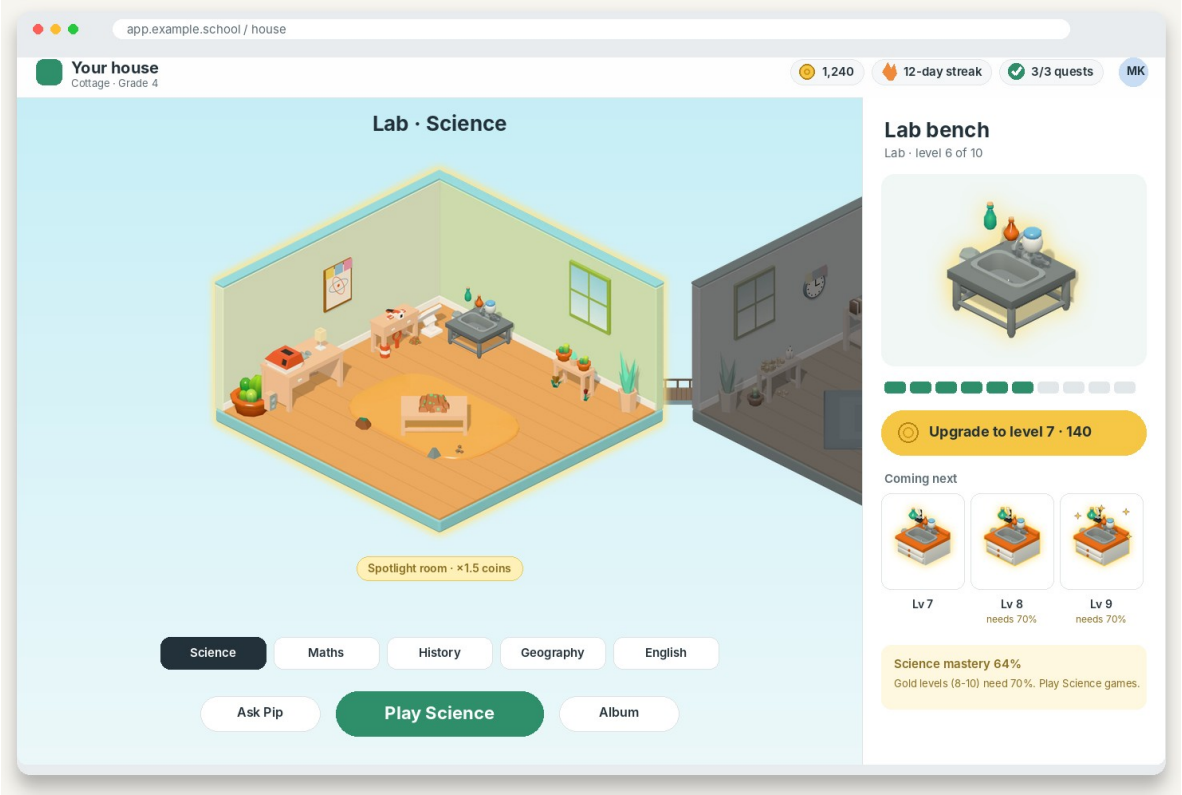
Home, Lab centred. The Spotlight room (weakest subject) pays ×1.5 coins.



Dragging from the Lab to the Maths workshop: both rooms fade as they move off centre.

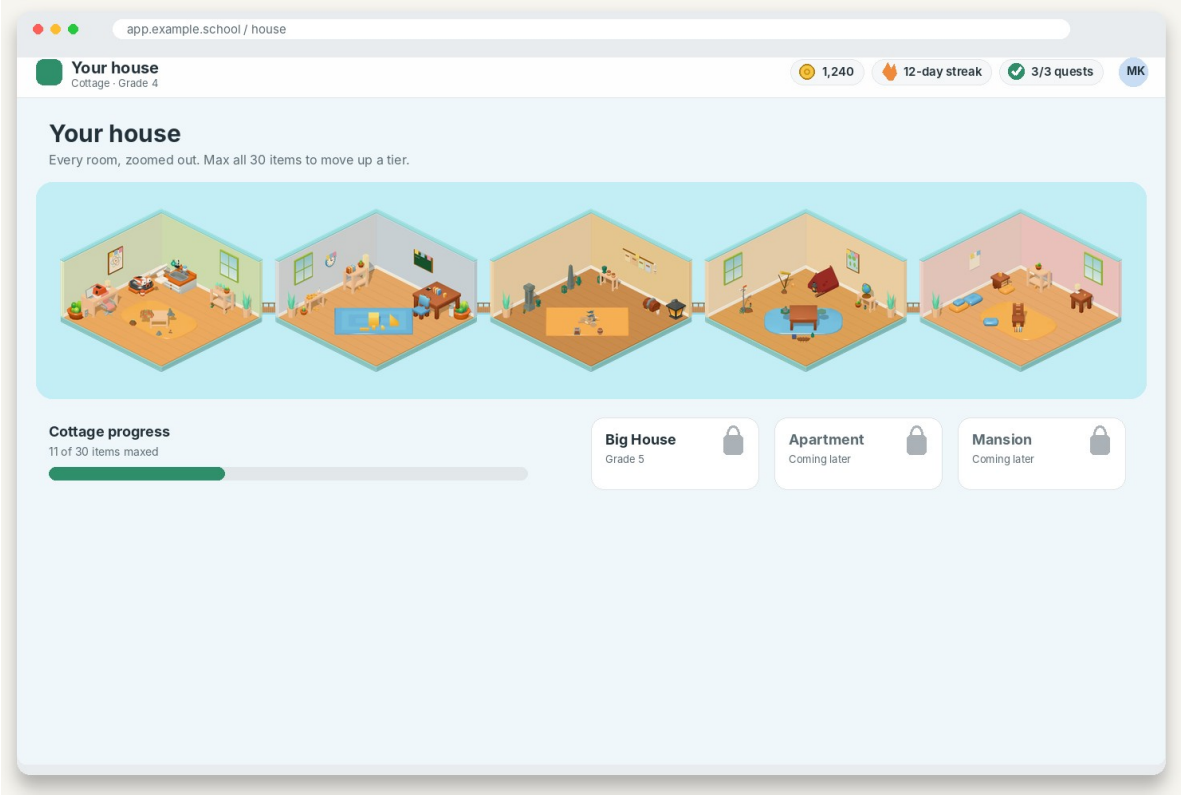
- The five rooms of the current tier sit side by side in one horizontal strip, joined by small wooden walkways. The student **drags sideways** (or taps a subject tab) to move along it.
- **Snap:** on release the strip springs (~250 ms) so the nearest room is centred.
- **Focus:** the centred room is full colour with a soft yellow glow. Every other room and the walkways are dimmed with one CSS filter: `saturate(1 − 0.8·d) brightness(1 − 0.45·d)`, d = distance from centre in rooms, capped at 1. During a drag d is fractional, so neighbours fade smoothly.
- Top bar: app logo, house and grade, coins, streak, quests, avatar. Under the house: subject tabs, then Ask Pip · **Play <subject>** · Album.
- The page opens centred on the Spotlight room. Narrow screens (phones) show the same strip at a smaller scale and the item panel as a bottom sheet.

1.2 Tapping an item and upgrading



Item panel: current level, upgrade cost, the next three looks and the mastery gate.

- Tapping an item in the centred room opens the side panel. Hit-test the item images front to back (layout.json order reversed) on their alpha channel, so clicks on transparent corners fall through.
- The panel shows the item, ‘level n of 10’, a 10-segment bar, the upgrade button with the next cost, the next three looks, and the subject mastery when a gold level is next.
- Upgrade calls the backend. The server checks coins and the gate and returns the new level. The page swaps that one image and plays a pop and sparkles. If the gate is not met the button reads ‘Needs 70% in Science’ and opens that subject's games.



Whole house and tier progress.

1.3 The 10-level pattern (same for every item)



Example: Lab bench. Every strip in this document has the same layout.

Level	What changes	Cost
1	Stage 1 (basic)	free
2	Add-on A (small prop)	40
3	Add-on B	60
4	Stage 2 (better object)	80
5	Add-on C	100
6	Warm glow (item panel only)	120
7	Stage 3	140
8	Gold rim (item panel only) · needs 70%	160
9	Sparkles (also in the room) · needs 70%	180
10	Stage 4 (best) · needs 70%	200

Add-ons stay once earned. In the room only sparkles are drawn; glow and gold rim show in the item panel, so the room stays readable.

2. How to build it

2.1 The art is already baked

Every room and item level in sections 3 and 4 is already rendered into web-ready images. The app loads one background per room and one image per item per level, and places them at the pixel positions in `layout.json`.

Folders

```
HackAssets/07_web_rooms/                copy to web/public/rooms/
  tier1/                                Cottage, Grade 4
    science/
      shell.webp                        floor, walls, windows, rug, plants (no upgradable items)
      layout.json                       position of every item image at every level
      lab_bench/lv01.webp ... lv10.webp
      grow_shelf/ space_corner/ robot_desk/ rock_table/ atom_poster/
    maths/ history/ geography/ english/
  tier2/                                Big House, Grade 5, same structure
HackAssets/08_design_v3/
  Design.pdf  catalog.json  images/     catalogue = recipe of every item (which 3D models make each stage)
HackAssets/09_bake_tool/                the script that makes the images (Node + three.js + Python)
```

layout.json

```
{
 "tier": 1, "room": "science", "name": "Lab", "size": [1252, 1026],
 "items": [                                          back to front
  {"id": "atom_poster", "name": "Atom poster", "slot": "wl", "z": 0,
   "levels": [{"level": 1, "file": "atom_poster/lv01.webp", "x": 327, "y": 279, "w": 88, "h": 143}, ...]},
  {"id": "lab_bench", "name": "Lab bench", "slot": "r1", "z": 1,
   "levels": [{"level": 1, "file": "lab_bench/lv01.webp", "x": 558, "y": 310, "w": 335, "h": 268}, ...]}
 ]
}
```

RoomView (React sketch). Scale the whole room with a CSS transform to fit the screen.

```
export function RoomView({ layout, levels, onTap }: RoomProps) {
  return (
    <div className="relative" style={{ width: layout.size[0], height: layout.size[1] }}>
      <img src={`${layout.base}/shell.webp`} className="absolute inset-0" alt="" />
      {layout.items.map(it => {
        const l = it.levels[(levels[it.id] ?? 1) - 1]
        return <img key={it.id} src={`${layout.base}/${l.file}`} alt={it.name} onClick={e => onTap(it.id, e)
  }
                    className="absolute" style={{ left: l.x, top: l.y, width: l.w, height: l.h }} />
      })}
    </div>
  )
}
```

2.2 Stack

Part	Use
React 19 + TypeScript + Vite 8	App
Tailwind CSS 4 + shadcn/ui	Panels, tabs, buttons, dashboards
Motion 14	House drag and snap, upgrade pop, coins flying
TanStack Query 5 / Router	API data, routes
openapi-typescript	Types generated from the FastAPI backend
vite-plugin-pwa	Install on school tablets

2.3 Data the app keeps (server side)

Table / field	Meaning
student_items(student_id, item_id, level)	Missing row = level 1.
student_tier(student_id, tier)	Current house. Changes on tier-up or by teacher override (logged).
wallet_tx(student_id, amount, reason)	Coins; balance = sum. An upgrade inserts a negative row in the same transaction as the level change.
topic_mastery → subject mastery	Average of the subject's topics; used for the 70% gate.

2.4 Rules for the code

The hackathon is judged on how effective the code is. Short, clear, data-driven code wins. These rules apply to every file.

- **No decorative or separator comments.** Never write `//----------`, `// ======`, `/* ***** */`, `#######` or banner headers. They add nothing and waste tokens.
- Comment only the *why* when it is not obvious. No restating what a line does, no commented-out code, no TODO lists in code.
- Data-driven: rooms, items, levels and prices come from layout.json and the API. Never hard-code an item, room or price in UI code.
- One generic HouseStrip, RoomView and ItemPanel. No per-room or per-item components.
- Use the ids in this document exactly (rooms: science, maths, history, geography, english; items as listed).
- Small functions, focused files, no copy-paste. Delete unused code before committing.
- The server owns coins, levels and gates; the browser only displays them and sends requests.
- When asking an AI to write code from this document, include this section in the prompt.

2.5 Changing or adding art

Each room is one catalogue entry (`catalog.json`, or one file per room in the repo). A stage is a list of parts: `PACK:model` plus flags (`h` height in metres, `w` width, `s` scale; `on`, `top`, L, R, F, B place it on, above, left, right, in front of or behind the base). Edit, run the bake tool for that room, commit the new images. The app code does not change.

3. Tier 1 · Cottage (Grade 4)

6 × 6 m rooms. 30 upgradable items. Full house = 32,400 coins. Room themes follow the Grade 4 topics.



Whole house, every item at level 1 (a new student).



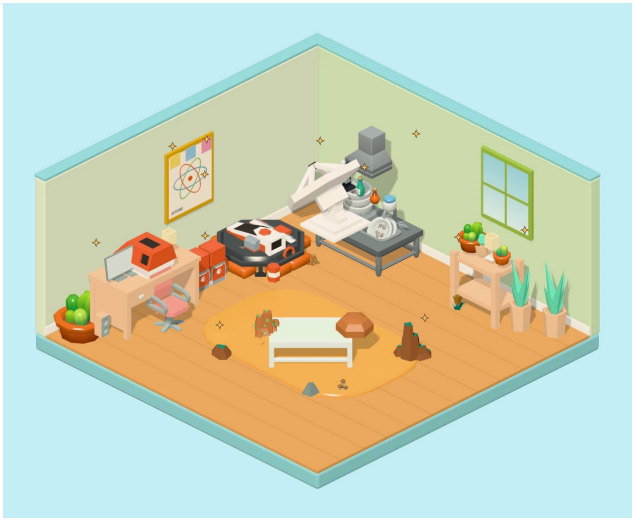
Whole house, every item at level 10 (end of the school year).

3.1 Lab · Science

Room id `science` · topics: food chains · states of matter · plants · walls #dff0d8 / #cfe8c6 · floor #d9a46a



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Lab bench `lab_bench`	right wall, back	kitchentable	kitchentable sink + bottle a green	kitchencounter sink backsplash + bottle c green, microscope*	kitchentable sink large decorated + microscope*, bottle a labeled green, extractorhood	bottle b brown / jar a small / trashcan
Grow shelf `grow_shelf`	right wall, front	cactus small	side table + cactus small, plant small1	bookcase open low + plant small1, plant small2, plant small3	bookcase open low + cactus medium, plant small3, lamp square table, potted plant	flower yellow a / cactus small / flower red a
Space corner `space_corner`	left wall, back	side table + meteor half	side table drawers + space craft1, satellite dish	landingpad small + space craft3, satellite dish	landingpad large + space craft5, satellite dish large, astronaut	rocks small ore / barrel / pipe stand
Robot desk `robot_desk`	left wall, front	table + robot	desk + robot, computer keyboard	desk + computer screen, robot, chair desk	desk + computer screen, laptop, robot, chair desk, astronaut	computer mouse / lamp square table / speaker small
Rock collection `rock_table`	centre	rock small a	table coffee + rocks ore	table coffee square + rocks ore, stone tall a	table coffee glass + rocks tall ore, meteor half, rock tall a	rock small b / stone small a / mushroom tan group
Atom poster `atom_poster`	on left wall	poster (atom)*	poster (atom)*	poster (atom)*	poster (atom)*	sticky note* / sticky note* / sticky note*

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Lab · item upgrades

Lab bench · Lab

id: lab_bench slot: r1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Grow shelf · Lab

id: grow_shelf slot: r2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Space corner · Lab

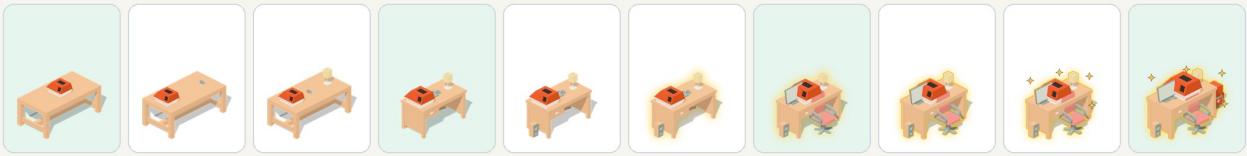
id: space_corner slot: l1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Robot desk · Lab

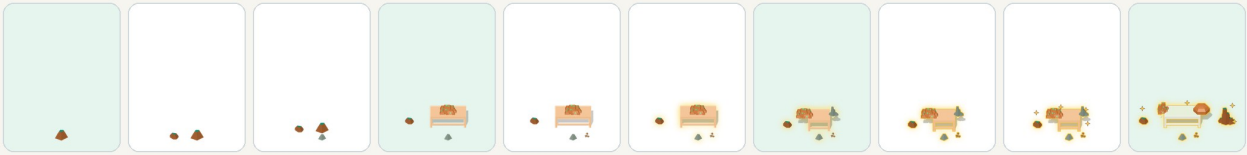
id: robot_desk slot: l2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Rock collection · Lab

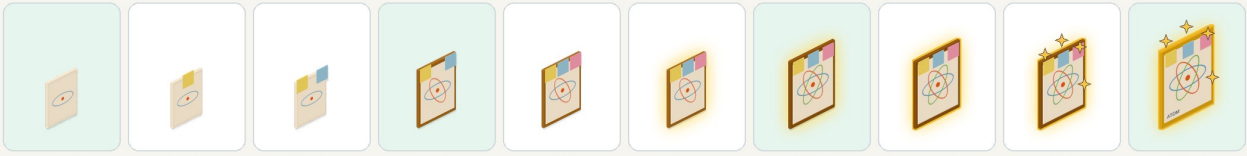
id: rock_table slot: c tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Atom poster · Lab

id: atom_poster slot: wl tier 1



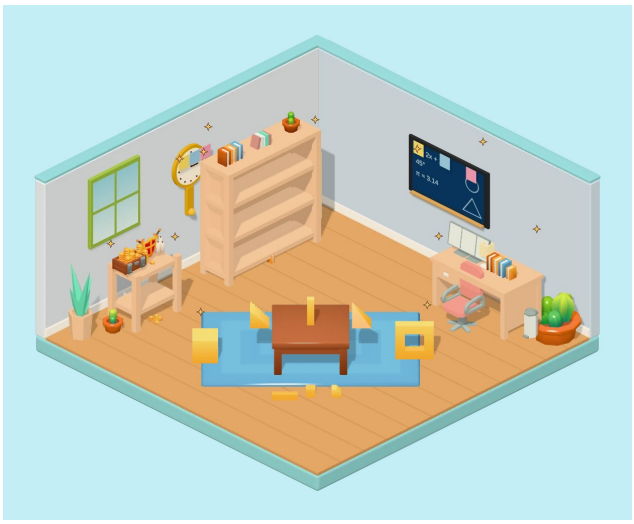
Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

3.2 Maths workshop · Mathematics

Room id `maths` · topics: multiplication · fractions · area · walls #dbe8f5 / #c9dcef · floor #cfa06e



All items level 1



All items level 10

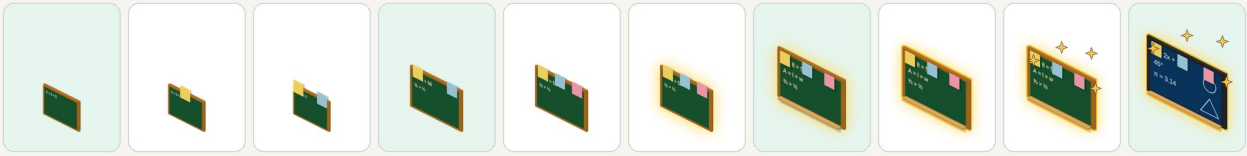
Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Chalkboard `chalkboard`	on right wall	chalkboard*	chalkboard*	chalkboard*	chalkboard*	sticky note* / sticky note* / sticky note*
Shape blocks `shape_blocks`	centre	primitive cube small	primitive cube + primitive slope	table coffee square + primitive cube small, primitive pillar, primitive slope outer corner	table medium + primitive stairs, primitive pillar, primitive slope outer corner, primitive window, primitive cube	primitive cube small / primitive slope half / primitive beam
Maths desk `maths_desk`	right wall, front	table small + chair	table medium + chair, calculator*	desk + chair desk, computer screen, calculator*	desk + chair desk, computer screen, computer screen, lamp square table	book set / book single / trashcan
Wall clock `wall_clock`	on left wall	clock*	clock*	clock*	clock*	sticky note* / sticky note* / sticky note*
Maths books `maths_books`	left wall, back	bookcase open low	bookcase open low + book set	bookcase open + book set	bookcase closed wide + book set, books	cactus small / book single / lamp square floor
Trophy shelf `trophy_shelf`	left wall, front	side table + coin stack small	side table + coin stack medium, coin stack small	cabinet medium + coin stack large, chest gold	bookcase open low + chest gold, coin stack large, sword shield gold	coin stack small / candle triple / cactus small

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Maths workshop · item upgrades

Chalkboard · Maths workshop

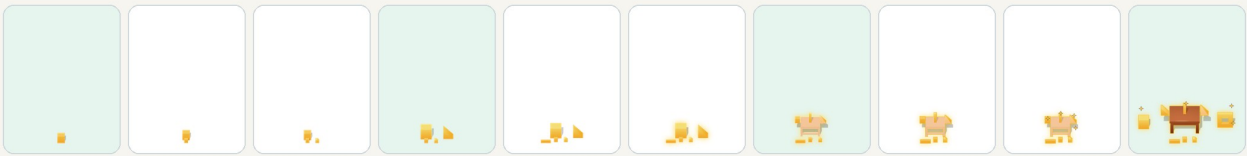
id: chalkboard slot: wr tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Shape blocks · Maths workshop

id: shape_blocks slot: c tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Maths desk · Maths workshop

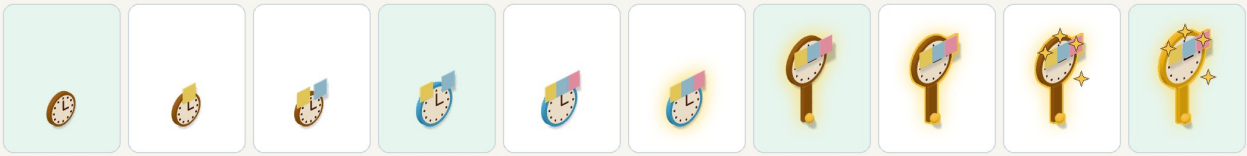
id: maths_desk slot: r2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Wall clock · Maths workshop

id: wall_clock slot: wl tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Maths books · Maths workshop

id: maths_books slot: l1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Trophy shelf · Maths workshop

id: trophy_shelf slot: l2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

3.3 Museum · History

Room id `history` · topics: Ancient Egypt · Greece · Rome · walls #f3e2c2 / #ead3a8 · floor #b9875a



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Pottery shelf `pottery_shelf`	right wall, back	urn	side table + urn, jar c small	cabinet medium + urn, jar c large, jar d medium	bookcase open low + urn, jar d large, jar c large, urn	lantern candle / jar c small / detail chalice
Egypt corner `egypt_corner`	left wall, back	urn	pillar obelisk	pillar obelisk + altar stone, urn	statue obelisk + pillar obelisk, altar stone, coin stack medium	cactus short / lantern candle / detail bowl
Greek statues `greek_statues`	left wall, front	statue block	statue column	statue ring + statue column	statue head + column large, column large	urn / flower yellow a / lantern candle
Roman siege model `roman_siege`	centre	siege ram	siege ballista	siege catapult	siege trebuchet + knight red	barrel small / box small / shield red
Treasure chest `treasure_chest`	right wall, front	box small	trunk medium	trunk large	chest gold + coin stack large, coin stack medium	coin stack small / candle triple / key
Timeline wall `timeline`	on right wall	timeline*	timeline*	timeline*	timeline*	sticky note* / sword / shield blue

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Museum · item upgrades

Pottery shelf · Museum

id: pottery_shelf slot: r1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Egypt corner · Museum

id: egypt_corner slot: l1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Greek statues · Museum

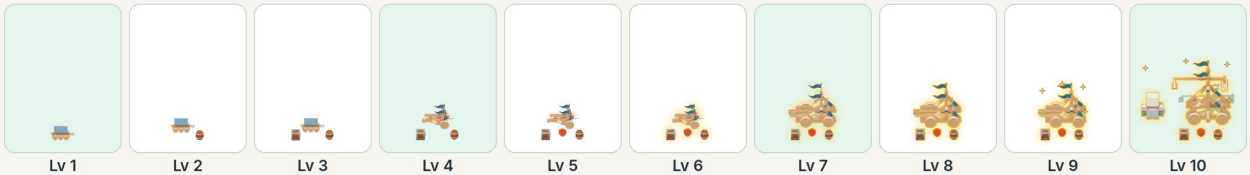
id: greek_statues slot: l2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Roman siege model · Museum

id: roman_siege slot: c tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Treasure chest · Museum

id: treasure_chest slot: r2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Timeline wall · Museum

id: timeline slot: wr tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

3.4 Explorer's den · Geography

Room id geography · topics: continents · rivers & mountains · weather & climate · walls #f6e7cf / #efdcbc · floor #c99a63



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
World map `world_map`	on right wall	poster (map)*	poster (map)*	poster (map)*	poster (map)*	sticky note* / sticky note* / sticky note*
Globe `globe`	right wall, front	side table + globe*	side table + globe*	table small + globe*, book set	globe* + book set	book single / cactus small / flower purple a
Landscape model `landscape_table`	centre	table coffee square + sand	table medium + grass hill, water	table medium long + river straight, grass forest, water island	table medium long + stone mountain, river straight, building village, grass forest	unit tree / unit boat / unit house
Telescope `telescope`	left wall, back	telescope*	telescope*	telescope* + book set	telescope* + book set, lantern standing	book single / rock small a / sticky note*
Weather station `weather_station`	left wall, front	weather station*	weather station*	weather station* + solarpanel	weather station* + solarpanel, flower yellow a	flower red a / grass / rock small c
Explorer camp `explorer_camp`	right wall, back	log stack	tent small open	tent small open + campfire stones	tent detailed open + campfire logs, canoe	log / mushroom red group / lantern standing

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Explorer's den · item upgrades

World map · Explorer's den

id: world_map slot: wr tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Globe · Explorer's den

id: globe slot: r2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Landscape model · Explorer's den

id: landscape_table slot: c tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Telescope · Explorer's den

id: telescope slot: l1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Weather station · Explorer's den

id: weather_station slot: l2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Explorer camp · Explorer's den

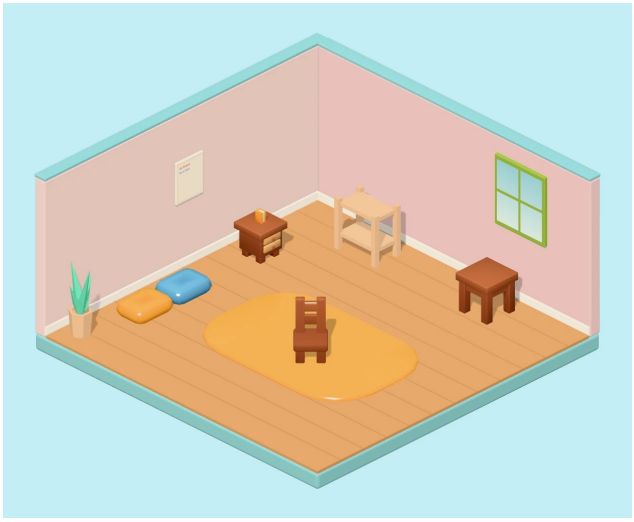
id: explorer_camp slot: r1 tier 1



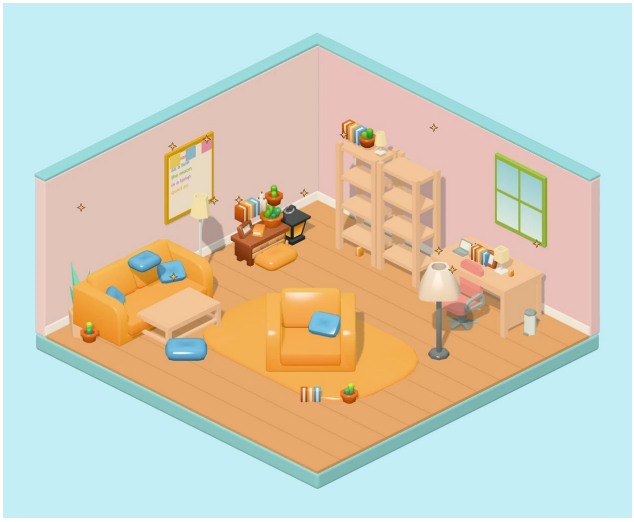
Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

3.5 Reading nook · English

Room id `english` · topics: parts of speech · tenses · paragraphs · walls #f4dfe4 / #ecd0d8 · floor #d4a274



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Big bookshelf `bookshelf`	right wall, back	bookcase open low	bookcase open low + book set	bookcase open + books	bookcase open + bookcase open, book set	cactus small / book single / lamp round table
Reading chair `reading_chair`	centre	chair a wood	chair	armchair pillows	armchair pillows + lamp standing	pillow / book set / cactus small
Writing desk `writing_desk`	right wall, front	table small	table medium + chair b wood, book single	desk + chair desk, laptop, book set	desk + chair desk, laptop, candle triple, book set, pictureframe standing	lamp square table / book single / trashcan
Story cabinet `story_cabinet`	left wall, back	cabinet small + book single	cabinet small decorated + book set	cabinet medium decorated + book set, candle	cabinet medium decorated + book set, candle triple, lantern standing	pillow / book single / cactus small
Cozy sofa `cozy_sofa`	left wall, front	pillow + pillow	bench cushion low	couch	couch pillows + lamp round floor, table coffee	pillow / book single / cactus small
Story wall `story_wall`	on left wall	poster (words)*	poster (words)*	poster (words)*	poster (words)*	sticky note* / sticky note* / sticky note*

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Reading nook · item upgrades

Big bookshelf · Reading nook

id: bookshelf slot: r1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Reading chair · Reading nook

id: reading_chair slot: c tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Writing desk · Reading nook

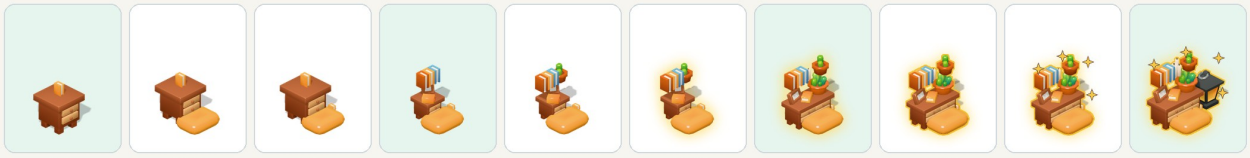
id: writing_desk slot: r2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Story cabinet · Reading nook

id: story_cabinet slot: l1 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Cozy sofa · Reading nook

id: cozy_sofa slot: l2 tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Story wall · Reading nook

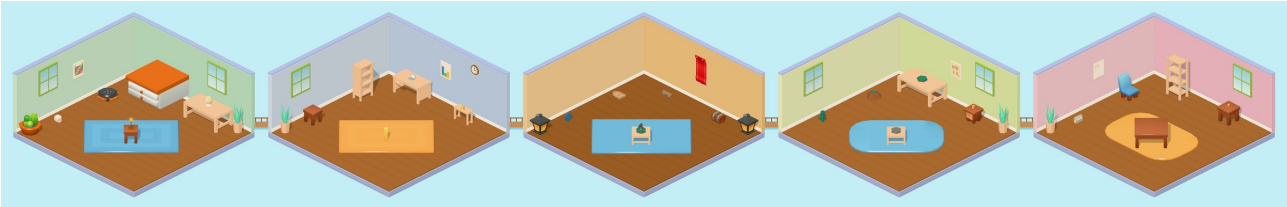
id: story_wall slot: wl tier 1



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

4. Tier 2 · Big House (Grade 5)

7 × 7 m rooms. 30 upgradable items. Full house = 32,400 coins. Room themes follow the Grade 5 topics.



Whole house, every item at level 1 (a new student).



Whole house, every item at level 10 (end of the school year).

4.1 Lab · Science

Room id `science` · topics: body systems · solar system · electricity · walls #c9ead9 / #b5e0cb · floor #9c6b45



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Fume-hood bench `fume_bench`	right wall, back	kitchencounter straight	kitchencounter straight a decorated + bottle a green	kitchencounter sink backsplash + kitchencounter straight b backsplash, microscope*, extractorhood	kitchencounter sink backsplash + kitchencounter straight decorated, microscope*, bottle a labeled green, extractorhood	jar a medium / bottle c green / trashcan
Solar system model `solar_system`	centre	table small + solar-system model*	table small + solar-system model*	table medium + solar-system model*	table medium + solar-system model*	meteor half / satellite dish / globe*
Rocket launch pad `rocket_pad`	left wall, back	landingpad small	landingpad small + space craft1	landingpad large + space craft3, satellite dish	landingpad large + lander, satellite dish large, astronaut	barrel / rocks small ore / robot
Human body model `body_model`	left wall, front	skull	table small + skull	skeleton + skull	skeleton + cabinet medium, ribcage	jar b small / bottle b green / book set
Circuit bench `circuit_bench`	right wall, front	table + lamp square table	desk + console, lamp square table	desk + console screen, console, chair desk	desk + console screen, computer screen, console, chair desk, speaker	computer keyboard / solarpanel / speaker small
Periodic table `periodic_table`	on left wall	poster (periodic)*	poster (periodic)*	poster (periodic)*	poster (periodic)*	sticky note* / sticky note* / sticky note*

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Lab · item upgrades

Fume-hood bench · Lab

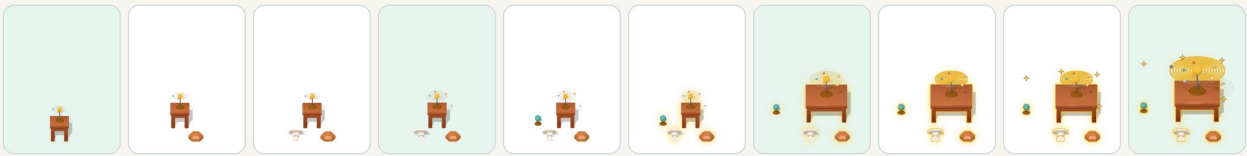
id: fume_bench slot: r1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Solar system model · Lab

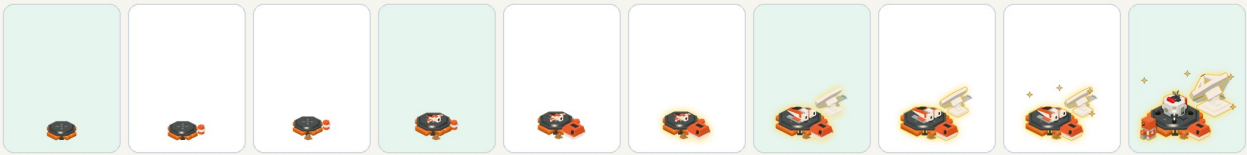
id: solar_system slot: c tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Rocket launch pad · Lab

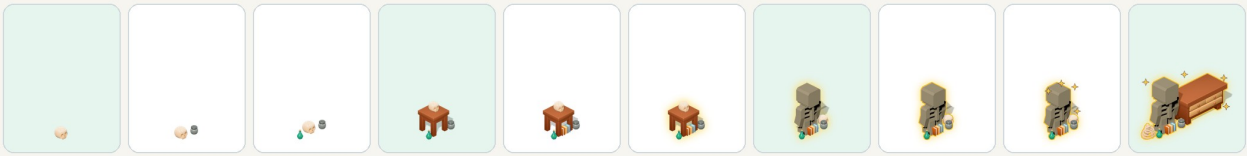
id: rocket_pad slot: l1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Human body model · Lab

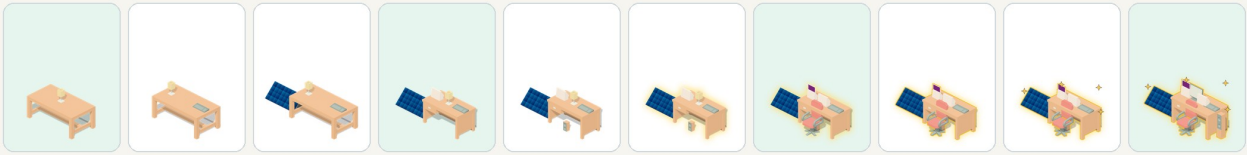
id: body_model slot: l2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Circuit bench · Lab

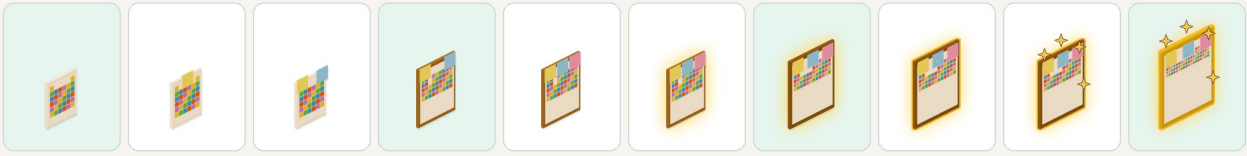
id: circuit_bench slot: r2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Periodic table · Lab

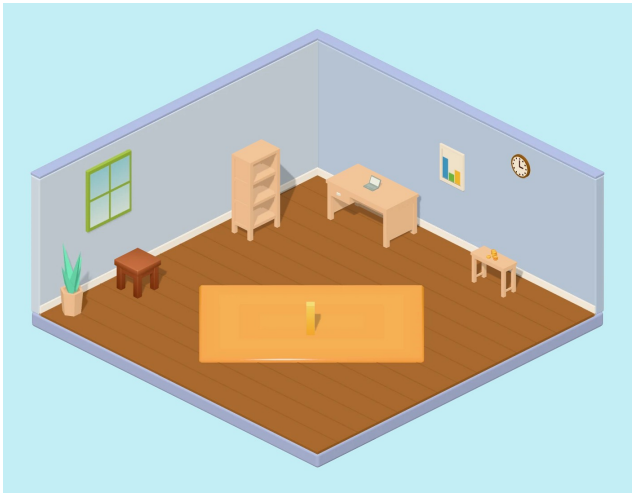
id: periodic_table slot: wl tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

4.2 Maths workshop · Mathematics

Room id `maths` · topics: decimals · angles & shapes · data & graphs · walls #cfe0f7 / #b9d0f0 · floor #9c6b45



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Data & graphs wall `data_wall`	on right wall	poster (graph)*	poster (graph)*	poster (graph)*	poster (graph)*	sticky note* / sticky note* / sticky note*
Geometry tower `geometry_tower`	centre	primitive pillar	primitive pillar + primitive slope	primitive stairs + primitive pillar, primitive slope half	primitive stairs + primitive pillar, primitive slope outer corner, primitive window	primitive cube small / primitive beam / primitive floor hole
Computer station `computer_station`	right wall, back	desk + laptop	desk + computer screen, computer keyboard	desk + computer screen, computer screen, chair desk	desk + computer screen, computer screen, laptop, chair desk, speaker	computer mouse / lamp square table / speaker small
Puzzle cabinet `puzzle_cabinet`	left wall, back	bookcase closed	bookcase closed doors	bookcase closed wide + primitive cube small	bookcase closed wide + primitive cube small, primitive slope outer corner, primitive pillar	cactus small / book set / lamp square floor
Board game table `game_table`	left wall, front	table small	table medium + primitive cube small	table medium + primitive cube small, coin stack small, chair	table medium long + primitive cube small, coin stack medium, chair, chair	book single / cactus small / lamp round table
Trophy cabinet `trophy_cabinet`	right wall, front	side table + coin stack small	cabinet medium + chest gold	bookcase open + chest gold, coin stack large	bookcase open + sword shield gold, chest gold, coin stack large	coin stack small / candle triple / cactus small

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Maths workshop · item upgrades

Data & graphs wall · Maths workshop

id: data_wall slot: wr tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Geometry tower · Maths workshop

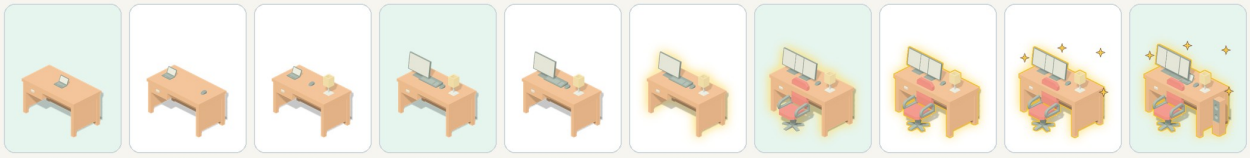
id: geometry_tower slot: c tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Computer station · Maths workshop

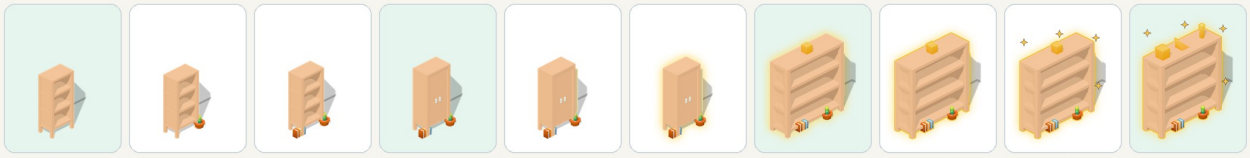
id: computer_station slot: r1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Puzzle cabinet · Maths workshop

id: puzzle_cabinet slot: l1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Board game table · Maths workshop

id: game_table slot: l2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Trophy cabinet · Maths workshop

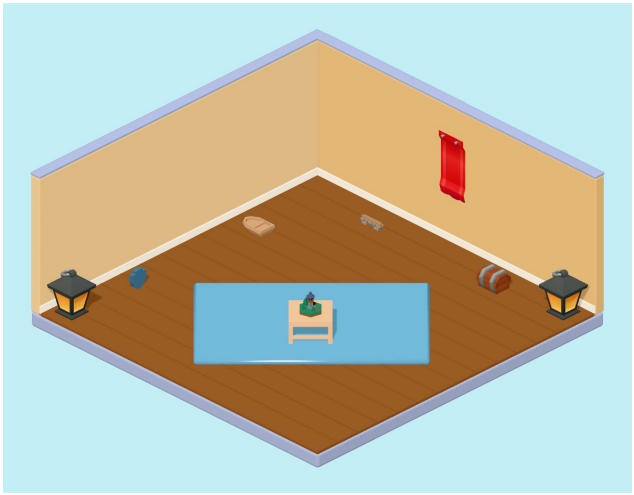
id: trophy_cabinet slot: r2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

4.3 Museum · History

Room id `history` · topics: castles · age of exploration · industrial revolution · walls #efd3a6 / #e5c48f · floor #8e5e3c



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Castle model `castle_model`	centre	table coffee square + building tower	table medium + building castle	table medium long + building castle, building village	table medium long + building castle, building village, building wall, flag blue	unit tower / unit wall tower / shield blue
Steam train `steam_train`	right wall, back	train wagon flat	train locomotive + train tender	table medium long + train locomotive, train wagon	table medium long + train locomotive, train tender, train wagon, lightpost	box small / barrel small / present
Explorer ship `explorer_ship`	left wall, back	boat small	side table + ship light	cabinet medium + ship light	cabinet medium + ship dark, cannon, chest	bottle / sword / globe*
Knight hall `knight_hall`	left wall, front	shield blue	knight blue	knight blue + shield blue, sword	king + knight red, knight blue	shield red / lantern standing / sword shield
Museum vault `museum_vault`	right wall, front	trunk small	trunk medium + pillar decorated	chest gold + pillar decorated, pillar decorated	chest gold + pillar decorated, pillar decorated, coin stack large	candle triple / urn / key
Banner wall `banner_wall`	on right wall	banner red	banner red + banner thin blue	banner shield red + banner thin blue, banner thin red	banner shield red + banner triple blue, banner triple red	shield red / sword / shield blue

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Museum · item upgrades

Castle model · Museum

id: castle_model slot: c tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Steam train · Museum

id: steam_train slot: r1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Explorer ship · Museum

id: explorer_ship slot: l1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Knight hall · Museum

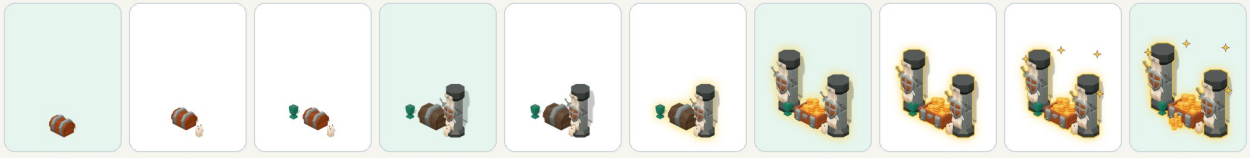
id: knight_hall slot: l2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Museum vault · Museum

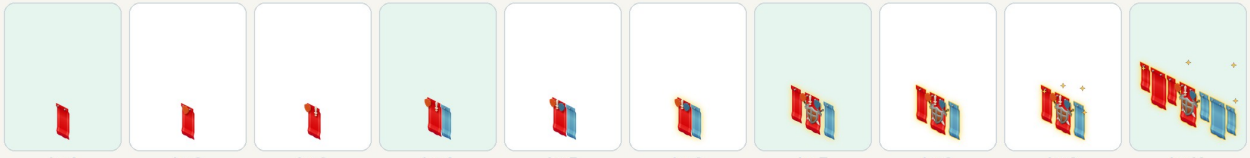
id: museum_vault slot: r2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Banner wall · Museum

id: banner_wall slot: wr tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

4.4 Explorer's den · Geography

Room id geography · topics: grid maps · natural resources · volcanoes · walls #e3efc9 / #d4e6b2 · floor #9c6b45



All items level 1



All items level 10

Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Grid-reference map `grid_map`	on right wall	poster (map)*	poster (map)*	poster (map)*	poster (map)*	sticky note* / sticky note* / sticky note*
Volcano model `volcano_model`	centre	table coffee square + stone	table medium + stone hill	table medium + stone mountain, sand rocks	table medium long + stone mountain, stone rocks, water rocks	rock small a / cactus short / rock small c
Resource map table `resource_table`	right wall, back	table + grass	table + dirt lumber, water	table medium long + building mine, building mill	table medium long + building mine, building mill, building water, solarpanel	unit mill / log stack / solarpanel
Atlas & compass stand `atlas_stand`	right wall, front	cabinet small + book single	cabinet small + book set, globe*	cabinet medium + book set, globe*	cabinet medium decorated + globe*, book set, lantern standing	book single / cactus small / flower purple a
Rock & fossil cabinet `rock_cabinet`	left wall, back	rock small c	side table + rock large a	bookcase open low + stone large a, rocks ore, bone	bookcase open + rocks tall ore, stone tall b, skull, rock large b	rock small b / stone small b / bone
Climate-zone garden `climate_garden`	left wall, front	cactus short	cactus short + cactus tall	tree palm short + cactus tall, tree pine small a	tree palm tall + cactus tall, tree pine small a, tree small fall	flower red a / mushroom red group / flower yellow a

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Explorer's den · item upgrades

Grid-reference map · Explorer's den

id: grid_map slot: wr tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Volcano model · Explorer's den

id: volcano_model slot: c tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Resource map table · Explorer's den

id: resource_table slot: r1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Atlas & compass stand · Explorer's den

id: atlas_stand slot: r2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Rock & fossil cabinet · Explorer's den

id: rock_cabinet slot: l1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Climate-zone garden · Explorer's den

id: climate_garden slot: l2 tier 2



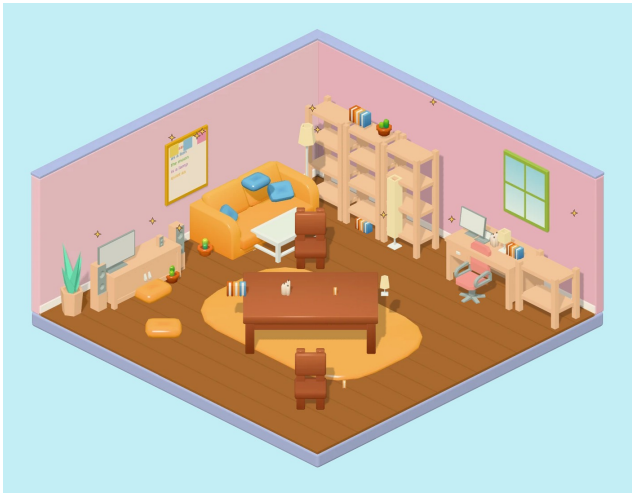
Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

4.5 Reading nook · English

Room id `english` · topics: similes & metaphors · fact & opinion · story structure · walls #f0cfe0 / #e6bcd2 · floor #9c6b45



All items level 1



All items level 10

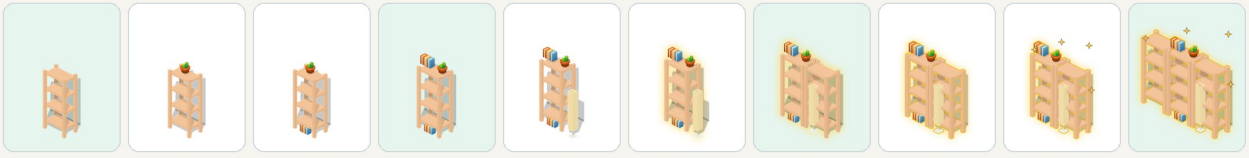
Item (id)	Where	Stage 1 · Lv 1	Stage 2 · Lv 4	Stage 3 · Lv 7	Stage 4 · Lv 10	Add-ons Lv 2 / 3 / 5
Library wall `library_wall`	right wall, back	bookcase open	bookcase open + book set	bookcase open + bookcase open, book set	bookcase open + bookcase open, bookcase open, book set	cactus small / book set / lamp square floor
Reading table `reading_table`	centre	table medium	table medium long + book set	table medium long + book set, candle, chair b wood	table medium long + book set, candle triple, book single, chair b wood, chair b wood	pillow / book single / lamp round table
Lounge sofa `lounge_sofa`	left wall, back	chair	couch	couch pillows	couch pillows + lamp round floor, table coffee glass	pillow / book single / cactus small
Author desk `author_desk`	right wall, front	table small + book single	desk + laptop, chair b wood	desk + laptop, computer screen, chair desk	desk + computer screen, laptop, candle triple, chair desk, bookcase open low	lamp square table / book set / trashcan
Listening corner `listening_corner`	left wall, front	radio	side table + radio	cabinet television + television vintage	cabinet television doors + television modern, speaker, speaker	speaker small / pillow / cactus small
Simile & metaphor wall `metaphor_wall`	on left wall	poster (words)*	poster (words)*	poster (words)*	poster (words)*	sticky note* / sticky note* / sticky note*

* made in code (no free model exists). Models are KayKit or Kenney, all CC0; exact files are in catalog.json.

Reading nook · item upgrades

Library wall · Reading nook

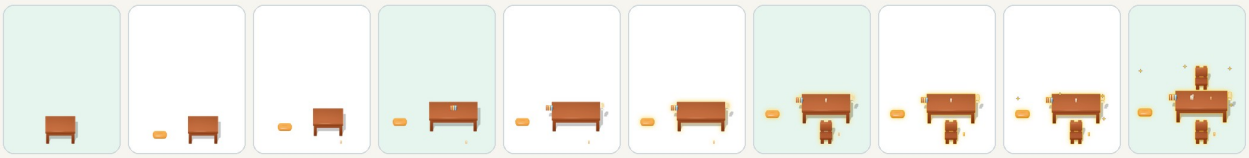
id: library_wall slot: r1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Reading table · Reading nook

id: reading_table slot: c tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Lounge sofa · Reading nook

id: lounge_sofa slot: l1 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Author desk · Reading nook

id: author_desk slot: r2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Listening corner · Reading nook

id: listening_corner slot: l2 tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

Simile & metaphor wall · Reading nook

id: metaphor_wall slot: wl tier 2



Lv 1	Lv 2	Lv 3	Lv 4	Lv 5	Lv 6	Lv 7	Lv 8	Lv 9	Lv 10
free	40 coins	60 coins	80 coins	100 coins	120 coins	140 coins	160 coins	180 coins	200 coins
STAGE 1	+ add-on A	+ add-on B	STAGE 2	+ add-on C	+ glow	STAGE 3	+ gold rim 70%	+ sparkles 70%	STAGE 4 70%

5. Assets and sources

Source	Used for	Licence
KayKit Furniture Bits	chairs, tables, armchairs, sofas, cabinets, rugs, cacti, frames, books	CC0
KayKit Dungeon Remastered	bottles, chests, coins, candles, banners, pillars, barrels, trunks	CC0
KayKit Restaurant Bits	lab counters, sinks, extractor hoods, jars	CC0
KayKit Prototype / Space Base / Halloween	geometry shapes, landing pads, solar panels, skulls, bones, lanterns	CC0
Kenney Furniture Kit (3D)	desks, bookcases, screens, laptops, lamps, plants, radios, TVs	CC0
Kenney Space / Castle / Pirate / Holiday kits	rockets, robot, astronaut, knights, siege machines, ships, steam train	CC0
Kenney Nature / Graveyard / Hexagon kits	statues, obelisks, rocks, tents, trees, urns, landscape and castle tiles	CC0
Inter, Pixelify Sans	UI fonts	SIL OFL

Made in code

No free model exists for these, so the bake tool builds them from simple shapes: globe, telescope, weather station, solar-system model, microscope, calculator, chalkboard, clock, posters (atom, periodic table, map, words, bar chart), timeline, window and sticky notes. They can be swapped for real models later by editing the catalogue.

Still missing (get manually)

- Pip the mascot (idle, talk, cheer, sleepy). Must be original art.
- Background music loops, one per room.
- Optional: a game-show set for Quiz Arena and art for the other game pages (not part of the house).